



Course Name: Artificial Intelligence Lab	Course Code: CSL604
Experiment No: <u>02</u>	Class: <u>TE</u>
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Date of Performance: _____	Date of Submission: _____

Aim: Assignments on State space formulation and PEAS representation for various AI applications.

Lab Experiment Outcome: At the end of the course, the students will be able to identify languages and technologies for artificial intelligence.

Mapping Table: (Note: 3: Highly Mapped, 2: Moderately Mapped, 1: Weakly Mapped)

LOs vs POs & PSOs Mapping Level						
POs	PO1:	PO2:	PO3:	PO4:	PO5:	PO6:
	PO7:	PO8:	PO9:	PO10:	PO11:	PO12:
PSOs	PSO1:			PSO2:		

Rubrics	Marks (_/03)
Attendance (3 Marks)	
Discipline (3 Marks)	
Short Oral (3 Marks)	
Correctness of Lab Report (3 Marks)	
Timely Completion of Lab Report (3 Marks)	

Total Marks (_/15) :

Subject In-charge Signature :



Experiment 2.

Aim: Assignment on state formulation & PEAS representation for various AI application.

Title: PEAS Representation of AI Image Generator Module

Primary Goal:

To generate high-quality images from user text prompts using Artificial Intelligence while maintaining accuracy, creativity, and user satisfaction.

Performance Measure.

- Accuracy of generated images according to the text prompt.
- Image quality and resolution
- Creativity and uniqueness of generated images
- Speed of image generation.
- User satisfaction and usability.
- Ability to understand natural language prompts
- Reduction of errors or unwanted content.
- System reliability and availability.
- Efficient use of computing resources

The system performance is measured by how correctly the AI understand the prompt and generates images quickly with good visual quality.

It also consider user satisfaction, system, stability and content safety.



Environment

The environment in which the AI image generator operates includes:

1. Users providing text prompts
2. Computers or mobile services
3. AI servers and cloud computing systems
4. Image generation models (like diffusion models)
5. Databases containing training images
6. Internet and communication networks
7. Software platforms or applications

Actuators

Actuators are the components through which the system performs actions

1. Image generation engine
2. Image rendering module
3. Image editing and enhancement system
4. Image saving or export module

Sensors

1. Text prompt input from users
2. User interface inputs (keyboard or touch)
3. Feedback from users about generated images
4. System logs and monitoring data

Conclusion

The PEAS model helps in understanding how an AI Image Generator works as an intelligent system by defining its PEAS.